

Benchmarking Traditional and Transformer-Based Models for News Topics Classification

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Abstract— Text classification is a common task in natural language processing and is widely used in news-related applications. News articles are produced every day in large quantities, which makes automatic categorization an important and practical problem. For many years, traditional machine learning models were the main solution for this task. Recently, transformer-based models have attracted attention due to their strong performance, but their behavior compared to older methods under different text representations is still worth studying.

In this study, we focus on news category classification and compare traditional machine learning models with transformer-based models using different types of text representations. The main purpose is to understand how the choice of representation affects classification results.

The experiments were conducted on a large news dataset using three different approaches. The first approach used TF-IDF features with classical classifiers. The second relied on word embeddings combined with neural and ensemble models. The third approach used an end-to-end DistilBERT model. All models were trained and evaluated using the same settings and evaluation metrics.

The results show that TF-IDF-based models achieved moderate accuracy, with SVM reaching 75.85% and MLP reaching 81.19%. Using FastText word embeddings improved performance, with MLP reaching 81.05%. Random Forest models consistently achieved around 66-69% accuracy across all representations. The best result was obtained by DistilBERT, which achieved an accuracy of 86.34%, showing the benefit of contextual representations for news classification.

Key words: Text Classification, Natural Language Processing (NLP), News Categorization, Machine Learning, Transformer Models, DistilBERT

1. INTRODUCTION

Text classification aims to enable automated processing and categorization of large-scale textual data. News comes online all the time, and sorting it by hand is just too slow. Automatic classification helps a lot. You can find info faster, get content suggestions, or just make decisions easier. People have tried many ways to do this. Old models used hand-picked features.[1] Newer ones, like deep learning and transformers, catch patterns and meaning in text much better than before.

Earlier studies mostly used simple features like TF-IDF combined with classical machine learning models,[1] such as support vector machines or ensembles. These methods are fast, easy to

understand, and usually work okay. Later, word embeddings were used. They show how words relate to each other and often improve results.[2, 3] Recently, transformer models appeared. They can look at the whole text and understand context,[4, 5] which usually makes them more accurate.

Even though these methods work, most studies focus on only one model or one representation. That makes it hard to know which works better in general. Not many studies compare TF-IDF, word embeddings, and transformers side by side. Without that, it's not clear what's best or if a method will perform well in real situations.[1, 2, 4, 6]

This study tries to fill that gap by comparing traditional machine learning models with

transformer-based models using different ways to represent text. We test three setups: TF-IDF with classical classifiers, word embeddings with neural and ensemble models, and an end-to-end DistilBERT model. All models are trained and tested the same way to keep things fair.[1, 2, 4, 6]

We do three main things in this paper. First, we put traditional, embedding-based, and transformer models next to each other. Second, we check how different text representations affect results. Third, we give some simple advice about balancing model complexity and performance. After this, Section 2 reviews earlier studies, Section 3 explains our methods, Section 4 shows results, and Section 5 wraps up with future ideas.

2. RELATED WORK

News headline categorization is a standard benchmark in NLP, where short, often ambiguous titles must be mapped into a predefined set of topical categories. The HuffPost News Category Dataset has become a common testbed due to its large scale and realistic challenges, including short-text sparsity, semantically overlapping labels, and class imbalance, which collectively stress both representation learning and classifier robustness.[1, 2, 4, 6]

Table 1 summarizes representative studies on news classification, highlighting the datasets used, adopted methods, key

findings, strengths, and limitations of each approach.

Table 1: Summary of Related Work on News Classification

Paper	Method	Dataset	Findings	Strengths	Limitations
Fuks et al., CS229 (2018) [1]	TF-IDF + Logistic Regression / SVM + classical baselines	HuffPost News Category Dataset (subset)	Reports strong development performance for TF-IDF with linear classifiers; discusses effects of class weighting and imbalance	Clear baseline framing and practical error analysis.	Uses a reduced label set/subset; results depend on split and preprocessing choices
Bhor (2022) [4]	BERT baseline + GAN-BERT (semi-supervised)	HuffPost News Category Dataset	Proposes GAN-BERT to leverage unlabeled data; shows advantages when labeled data is limited	Contextual modeling; addresses low-label regimes	Compute/training-sensitive; outcomes vary with balancing and hyperparameters
Bogery et al., IJACSA (2019) [2]	Word2Vec + TF-IDF + (NB, SVM, KNN, GB)	HuffPost News Category Dataset	Multinomial Naive Bayes achieved 90.12% accuracy with approximately 90% recall. in their setup	Hybrid features effective for short headlines; broad classifier comparison.	Performance sensitive to feature engineering and data handling; limited standardization.
Yang et al., IJCAI (2020) [6]	Seeded Biterm Topic Models + embeddings	HuffPost-derived subsets (HuffN10, HuffN4)	Improves Macro-F1 on overlapping categories using biterm co-occurrence and embedding-informed.	Explicitly targets short-text sparsity and label overlap explicitly.	Relies on constructed subsets and weakly supervised assumptions.

A strong line of work treats the task as a supervised classification problem over sparse lexical features. Fuks et al.[1] evaluate classical baselines using bag-of-words and TF-IDF representations combined with models such as logistic regression and SVM. Their analysis highlights that TF-IDF with linear decision boundaries can be highly competitive, while also emphasizing how imbalance-aware strategies (e.g., weighting) change error patterns across frequent and rare classes, which is especially relevant when categories are unevenly represented.

To move beyond purely lexical matching, Bogery et

al.[2] explore hybrid feature engineering by incorporating Word2Vec-based semantic information alongside TF-IDF and then comparing multiple classifiers (e.g., Naive Bayes, SVM, KNN, Gradient Boosting). Their findings indicate that simple probabilistic models can perform strongly under this representation, reporting 90.12% accuracy and high recall for Multinomial Naive Bayes in their experimental setup. This suggests that, for short headlines, combining lexical salience (TF-IDF) with coarse semantics (embedding features) can reduce sparsity and improve separability for several categories.

Because many news labels are not cleanly separated (e.g., education-related labels that overlap with campus or policy terms), topic-model-based approaches have been investigated as a way to represent headlines with compact latent structure. Yang et al.[6] propose seeded biterm topic modeling that leverages word co-occurrence patterns (biterms) and embedding-informed priors, then evaluate on HuffPost-derived subsets designed to emphasize overlapping categories (HuffN10/HuffN4). Their results, reported using Macro-F1, show that modeling co-occurrence structure can be particularly beneficial when categories are correlated and the observed text is short.

More recent studies focus on contextual encoders that learn representations conditioned on surrounding words rather than fixed vectors per token.[5] Bhora [4] studies BERT-based classification on the same dataset and extends it with a semi-supervised GAN-BERT framework intended to exploit unlabeled data. The reported outcomes emphasize a practical setting where labeled data may be limited: contextual models offer strong representational capacity, and semi-supervised strategies can provide gains when annotation coverage is low, albeit at the cost of increased training complexity and sensitivity to hyperparameters.

Within these works, a main methodological opposition is efficiency versus representational expressivity. A linear model and TF-IDF represents a fast, powerful, and interpretable baseline. It can struggle with the same meaning or context. Word2Vec-augmented pipelines[2] add a layer of semantic similarity that can improve performance and generalisability on relatively short text data. The designs are still constrained by engineering choices as they remain mostly context-free. Ultimately, topic-model approaches[6] learn latent structure that explicitly models the sparsity of short text and overlapping labels. However, the evaluation primarily hinges on more costly “fully supervised” learning applied to a subset of the complete label set mainly on some assumptions and synthetically built labels. It is therefore impossible to make any claim about the effectiveness of these structure-aware methods in a truly “fully supervised” setting.

A recurring limitation in the literature is that results are not always directly comparable due to differences in label subsets, preprocessing pipelines, train/test splits, and reporting metrics. In response, this study adopts a controlled evaluation protocol on the HuffPost News Category Dataset by implementing representative approaches aligned with prior work—TF-IDF baselines, Word2Vec-based models, and topic-distribution features (LDA) coupled with classical classifiers—under a single consistent preprocessing, split strategy, and metric suite, enabling a fair comparison and a clearer

understanding of which representation family best suits short, multi-class news categorization.[1, 2, 4, 6]

3. METHODOLOGY

3.1 Data Collection

A cleaned corpus taken from the HuffPost News Category Dataset was used for the experiment. Initially, the dataset had several categories; however, for the purpose of this project, the labels were mapped into four single-label classes to make it a supervised multi-class (single-class per sample) classification task, as follows: OTHER (0), POLITICS (1), WELLNESS (2), and ENTERTAINMENT (3). the final dataset consisting of 136192 samples after cleaning and validation Every example contains one text document and exactly one associated category label.[4, 2, 1]

3.2 Data Preprocessing

effort to reduce noise while preserving semantic meaning. The pipeline included tokenization and lemmatization, removal of stop words, filtering tokens to retain only verbs, nouns, adjectives, and proper nouns, as well as the elimination of domain-specific terms (e.g., “*huffpost*”, “*click*”, “*email*”). In addition, extraneous characters and textual artifacts were removed. For classical machine learning models, the cleaned text was transformed into fixed-length numerical vectors suitable for statistical classification. For transformer-based training, tokenization[7] was performed using the pretrained DistilBERT tokenizer [4] to generate the required inputs, including token IDs and attention masks, in the expected format.

3.3 Feature Engineering

To compare different representation families under the same dataset, multiple feature schemes were used:

- TF-IDF is a type of high-dimensional sparse vector that captures term salience through frequency statistics.[1]
- FastText word embeddings: Document-level dense vectors built from pretrained FastText embeddings (size=100) focusing on the semantic similarity of documents, not just the overlap of words.[3]
- The LDA topic distributions are latent topic features that represent each document compactly as a probability distribution in the latent topic space. This feature is to safeguard the sparsity of short texts. Additionally, this feature encapsulates the thematic

structure in latent topics.[6, 8]

- DistilBERT contextual representations: end-to-end contextual encoding where token meaning is modeled relative to surrounding context for final classification.[4]

3.4.3.4 N-gram Detection and Bigram Creation

To capture common phrases in news articles (e.g., "White House", "Donald Trump"), a bigram detection step was applied using the Gensim Phrases model with a threshold of 10.[9] This improved the representation of compound terms in all feature types.

3.5 Data Splitting Strategy

The original class distribution is preserved across the stratified split. By setting a random seed of 42, the dataset was separated into 35% testing and 65% training. For transformer fine-tuning, the training section was split up again to form training and validation of training to monitor performance and select the best checkpoint.

4. MODEL DESIGN AND EXPERIMENTAL SETUP

4.1 Model Selection

The study assess classifiers from complementary methodological families.

- Support Vector Machine (SVM) is a strong classical baseline. Due to computational constraints with large datasets, the RBF kernel SVM was trained on only 10% of the TF-IDF training data.
- Random Forest (RF) uses an ensemble model which enables us to explore non-linear decision boundaries on dense representations.
- The Multilayer Perceptron (MLP) is a neural baseline that effectively learns non-linear patterns from dense feature vectors.
- DistilBERT is a lightweight transformer encoder that fine-tunes the evaluation of contextualized representations.

4.2 Model Architecture

Classical models receive fixed feature vectors (TF-IDF, Word2Vec document vectors, or LDA topic distributions) followed by the corresponding classifier. The MLP for TF-IDF features consisted of (128, 256, 128) hidden layers, while for word embeddings it used (128, 128) hidden layers, both with early stopping and dropout regularization. DistilBERT uses the pretrained distilbert-base-uncased encoder and a task-specific

classification head (fully connected layer with softmax) to output probabilities over the four single-label classes.

4.3 Model Training

All models were trained on the training split under consistent conditions. Classical models were fitted using standard solvers from widely used machine learning libraries. DistilBERT was fine-tuned for 3 epochs (early stopping) with batch size 16, learning rate $2e-5$, and seed 42. Training was stopped early due to convergence, with final training loss of 0.289. Model checkpoints were saved during training, and the best model was retained based on validation performance.

4.4 Hyperparameter Configuration

hyperparameters were selected with the goal of achieving good generalisation while avoiding unnecessary model complexity. For classical models, settings were chosen through a mix of GridSearch and empirical tuning. The SVM used a regularisation parameter of $C = 3$. The Random Forest model limited tree depth to 3, while the number of estimators ranged from 5 to 95 depending on the text representation.

The MLP was trained using the Adam optimiser with early stopping, and its hidden-layer structure was adjusted to match the input representation. For topic modelling, LDA was configured with 70 topics, selected based on coherence score comparison across candidate values [15, 20, 30, 35, 40, 45, 50, 60, 70].[8]

For transformer-based models, DistilBERT weights were retained and only task-specific layers were fine-tuned. Early stopping and dropout were applied where appropriate to reduce overfitting.

4.5 Evaluation Metrics

When evaluating the performance, we rely on the standard multi-class metrics of accuracy, precision, recall, and F1-score. Since the task is single-label (one true class per sample), metrics were computed in the standard multi-class setting. In addition, inference time per sample was recorded to compare computational efficiency across model families.

5. RESULTS AND DISCUSSION

Table 2 summarises how the different models performed across the various text representations, showing training accuracy, testing accuracy, and inference time per sample. As we can see, there are noticeable differences both in prediction quality and computational efficiency depending on the model and representation used.

Table 2: Performance Comparison Across Models and Representations

Model	Representation	Train	Test	Inference Time (s/sample)
SVM	TF-IDF	99.92%	75.85%	0.0039044
Random Forest	TF-IDF	66.17%	66.16%	0.0000009
MLP	TF-IDF	86.80%	81.19%	0.0000386
SVM	Word Embeddings	88.55%	79.51%	0.0026034
Random Forest	Word Embeddings	69.56%	69.49%	0.0000292
MLP	Word Embeddings	83.89%	81.05%	0.0000179
Linear SVM	LDA Topics	68.14%	68.34%	0.0000002
Random Forest	LDA Topics	66.16%	66.16%	0.0000006
MLP	LDA Topics	68.67%	68.49%	0.0000703
DistilBERT	Contextual	86.65%	86.34%	–

model	precision	Recall	F1-score
DistilBERT	0.8643	0.8634	0.8637

Looking at the overall performance, transformer-based modeling clearly leads the way.

DistilBERT achieved the highest test accuracy of **86.34%**, with training accuracy very close at **86.65%**, which suggests it generalises well and does not overfit. This confirms that using contextual embeddings that understand words in their surrounding context gives a real advantage for multi-class news classification.[4, 7]

Beyond overall accuracy, DistilBERT demonstrated consistently balanced performance across all evaluation metrics. The model achieved a precision of 86.43%, a recall of 86.34%, and an F1-score of 86.37%. These results indicate that DistilBERT not only predicts the correct class with high reliability, but also maintains an effective balance between false positives and false negatives across categories. This further confirms the robustness of contextual representations in handling multi-class news classification tasks

Among the classical machine learning models, the **MLP paired with TF-IDF features** performed the best, achieving **81.19%** test accuracy. It was closely followed by the MLP using word embeddings, which reached **81.05%**, showing that neural networks can make good use of richer representations, while simpler or linear models struggle more to capture the nuances of the text.[1]

Support Vector Machines gave mixed results. For instance, the **SVM with TF-IDF** had extremely high training accuracy (**99.92%**), but the test accuracy dropped to **75.85%**, clearly showing overfitting. Switching to word embeddings helped slightly, bringing test accuracy up to **79.51%**, which indicates that the input

representation has a big impact on generalisation.

Models based purely on **LDA topic representations** performed consistently lower, with test accuracy between **66% and 68%** across all classifiers. This tells us that while topic distributions capture some high-level structure, they are not enough on their own to differentiate news categories effectively.[2]

Finally, when we look at computational efficiency, classical models are extremely fast, often taking only microseconds per sample. **DistilBERT**, on the other hand, requires more processing time, but its improved accuracy demonstrates a reasonable trade-off between speed and performance. In other words, it costs a little more to compute, but the results are clearly better.

5.1 DISCUSSION

The performance differences mostly come from the features we used. TF-IDF just counts word frequency to see what's important. It doesn't really understand word meanings or context. That's why TF-IDF models only reach okay results.[1]

FastText word embeddings, which incorporate subword information, provided a robust semantic representation that improved over TF-IDF baselines. The MLP achieved 81.05% accuracy with FastText embeddings, demonstrating their effectiveness. However, compared to contextual embeddings like DistilBERT, FastText still lacks full contextual understanding..[3]

DistilBERT did the best. It looks at words in context and understands both meaning and grammar better than TF-IDF or embeddings. Compared to similar studies, its results are consistent and competitive, showing transformers are very effective for classifying news[4, 5, 7]

The strong and well-balanced precision, recall, and F1-score values further support the superiority of DistilBERT over classical and embedding-based approaches. Unlike TF-IDF or FastText representations, which may bias predictions toward dominant classes, DistilBERT maintains stable performance across categories, making it a reliable choice for real-world multi-class news classification scenarios.

6. CONCLUSION

This study compared old and new models for news classification using different ways to represent text. We wanted to see how the choice of features affects performance when the tests are .fair

We implemented four feature representations: TF-IDF, FastText word embeddings, LDA topic distributions, and contextual transformer embeddings. Among classical models, MLP with TF-IDF achieved 81.19%, while MLP with FastText reached 81.05%. The computational limitations of SVM with RBF kernel necessitated training on only 10% of data. DistilBERT achieved the highest accuracy at 86.34%, confirming the superiority of contextual representations.

These findings give some practical ideas about choosing models and features, and show the trade-offs between complexity and performance

7. FUTURE WORK

The results are promising, but there's more to explore. Future studies could try bigger and more varied news datasets, different languages, or even .live news streams

Other ideas include testing different transformers and tuning strategies to make them better.

Combining topic features with embeddings, or making models faster and lighter, could also help. Finally, trying these models in real news apps or recommendation systems would show how useful they are in practice

- Experiment with different word embedding architectures beyond FastText[5]
- Explore the impact of different bigram thresholds on classification performance
- Investigate hybrid models combining LDA topics with word embeddings[6]
- Optimize transformer inference time for real-time news categorization[4]

8. REFERENCES (IEEE STYLE)

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